

Variables:

- age
- total_cholesterol
- total_points
- smoker
- hdl
- sbp
- treated
- total_10_year_risk
- gender

age: Determines the user's age and state their points of initial likelihood of a heart attack

total_cholesterol: Depending on the age of the user and the amount of cholesterol the user has will give them additional points.

smoker: Boolean variable, if true then the user will gain additional points based on their age.

nonsmoker: Boolean variable, if false then the user will not gain any points regardless of their age.

hdl: if the user has an hdl value of 60 then the user will lose points, otherwise the user can either gain no or some points.

systolic_bp: depending on the value of systolic bp and the boolean variable whether it is treated or not, the user could gain 0-3 points.

point: it will be used as a stored variable, gathering points that will later on be used at the end.

total_10_year_risk: the variable points will be converted into percentages stored as the variable "total_10_year_risk" by using a conversion factor that we create. It will later be printed out as the final output for the user.

Gender: determines whether the user is male or female, will change the system and points

Test Cases on next page:

Case #	sex	age	total_cholesterol	hdl	sbp	smoker	treated	total_10_year_risk	Type
1	M	23	160	39	120	N	Y	<1%	typical
2	M	20	180	40	120	Y	Y	2%	Extreme low
3	M	20	200	50	150	N	N	<1%	Extreme low
4	F	20	205	55	160	Y	N	3%	Extreme low
5	M	38	185	54	162	Y	Y	8%	typical
6	F	76	168	58	132	N	N	11%	typical
7	M	34	198	60	143	Y	Y	1%	typical
8	F	62	195	70	174	N	Y	5%	typical
9	M	75	161	57	150	N	Y	20%	typical
10	M	45	191	43	135	N	N	4%	typical
11	F	43	198	31	170	Y	N	5%	typical
12	M	73	166	55	140	Y	N	10%	typical
13	F	29	165	46	165	N	Y	<1%	typical
14	F	20	160	60	133	Y	N	<1%	Extreme low
15	F	45	161	57	123	N	Y	1%	typical
16	M	54	164	48	132	N	N	6%	typical
17	M	79	187	46	143	Y	N	25%	typical
18	M	34	178	42	150	N	N	<1%	typical
19	M	62	169	55	123	N	Y	10%	typical
20	F	45	199	56	158	Y	Y	8%	typical
21	M	79	250	48	150	N	N	25%	typical
22	M	54	260	50	142	Y	Y	20%	typical
23	M	27	270	57	162	Y	N	6%	typical
24	F	34	280	43	175	N	N	1%	Extreme high
25	F	20	270	60	140	N	Y	<1%	Extreme low
26	M	20	160	44	148	Y	N	2%	Extreme low
27	F	24	250	44	125	Y	N	4%	typical
28	M	64	241	58	162	Y	Y	25%	typical
29	M	79	223	59	130	N	N	16%	typical
30	F	34	254	49	150	N	N	<1%	typical

